

Using Futures and Options to Value Real Investments

From Chapter 11 of *Valuation: The Art and Science of Corporate Investment Decisions*, 2/e.
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Chapter Overview

In this chapter, we use the market prices of traded derivative securities, such as options, forwards, and futures contracts, to value real investment opportunities. These prices allow us to convert risky future cash flows into their certainty equivalents, which can then be discounted using the risk-free rate.

The chapter describes a three-step valuation process that is closely related to the three-step discounted cash flow (DCF) process described in earlier chapters. The following grid summarizes the traditional DCF and the derivative securities approaches to valuation and reveals that the relevant difference between the approaches lies in how we adjust for risk. In the traditional DCF approach, we estimate expected cash flows and adjust for risk by using a risk-adjusted discount rate, whereas in the derivative securities approach, we use market prices of publicly traded securities to estimate certainty-equivalent cash flows, which are then discounted using the risk-free rate. An important advantage of the derivatives approach is that it uses market forecasts based on market prices, whereas the traditional DCF approach relies on management's own price forecasts.

	Traditional DCF Approach to Valuation	Derivative Securities Approach to Valuation
Step 1	Forecast the amount and timing of future cash flows.	Forecast the amount and timing of future cash flows.
Step 2	Estimate a risk-appropriate discount rate.	Use observed market prices to estimate certainty equivalents of expected future cash flows.
Step 3	Discount the investment's expected cash flows using the risk-adjusted discount rate to find its present value.	Discount the certainty equivalents of the cash flows of the investment using the risk-free rate to find its present value.

1 INTRODUCTION

Two important developments occurred in the spring of 1973 that had a profound effect on both the theory and practice of finance. The first was the April opening of the first organized exchange for trading stock options, the Chicago Board Options Exchange (CBOE).¹ The second development was the May publication of the Black-Scholes model, which provided a formula for valuing options.² In the ensuing decades, we have seen the proliferation of a variety of financial markets that trade options as well as other **financial derivatives**.

Financial derivatives are securities whose value is *derived from* the value of another security or asset. The latter is commonly referred to as the underlying asset, which could be shares of stock, a commodity, or any other asset.³ There are three principal types of derivative securities: options, forwards, and futures. An **option** contract provides the holder with the right, but not the obligation, to purchase (call option) or sell (put option) a specified asset (e.g., shares of common stock or a parcel of land) for a specified price (called the exercise or strike price) within a specified period of time. **Forward** contracts and **futures** contracts are similar to option contracts in that they also represent agreements to buy or sell an asset at a certain time in the future for a certain price. However, unlike options, which do not obligate the holder to exercise the contract—forward and futures contracts do. For a review of specific differences between forward and futures contracts, see the Technical Insight box two pages ahead. In addition, Appendix A reviews the basics of options for those who want a brief refresher.

The development of markets for futures, forwards, and options is having an important influence on how investment projects are valued in practice. In this chapter, we provide two examples to illustrate how this is done. The first example uses a technique commonly referred to as “pricing against the forward curve.” Here we use forward market prices to estimate the certainty equivalence of future cash flows. The second example uses option prices to help value investment opportunities that have option-like features. In this example, we show how option prices can be used to capture the value of a firm’s option to default on its debt obligations. Both of these examples illustrate the benefits of using market prices, whenever possible, when valuing risky investments.

Although the valuation approach used in this chapter is indeed somewhat different from the approaches used in earlier chapters, the similarities are really more important than the differences. In all cases, we are ultimately forecasting future cash flows that are discounted to the present; the difference comes from the method of coming up with the cash flows and the discount rate that is used. Moreover, using derivatives prices to value

¹ Derivative markets for futures contracts had long existed. For example, the Chicago Board of Trade (CBOT) was established in 1848 to allow farmers and merchants to contract for basic commodities.

² Fischer Black and Myron Scholes, “The Pricing of Options and Corporate Liabilities,” *Journal of Political Economy* 81 (May–June 1973): 637–59.

³ Swaps provide another important type of derivative contract that is widely used in hedging transactions; however, we will not utilize them in our examples. A **swap** is a derivative contract in which two counterparties exchange one stream of cash flows for another stream. These streams are called the “legs” of the swap. For example, take the case of a plain vanilla fixed-to-floating interest rate swap. Party A makes periodic interest payments to party B based on a variable interest rate, while party B makes periodic interest payments based on a fixed rate of interest. The payments are calculated using a notional amount. Swaps are over-the-counter (OTC) derivatives, which means that they are negotiated outside exchanges and cannot be bought and sold like securities or futures contracts, but are all unique.

cash flows is also directly analogous to the use of comparables. However, instead of using another business as the comparable to value a stream of future cash flows, we use derivative prices as the appropriate comparable to value individual cash flows.

The techniques that we focus on in this chapter were originally designed for commodity-based businesses like mining, basic chemicals, and oil and gas exploration, where the cash flow risks are derived from commodity prices for which there are publicly traded derivatives. For this reason, the examples we present in this chapter focus on extractive industries, with a particular focus on the oil and gas industry, where this approach is very popular.

Before we launch into the study of the derivatives approach to asset valuation, we want to emphasize a few important points that we hope will influence how you approach the material in this chapter:

- Because this approach is newer than traditional DCF analysis, there is a tendency to think that this material represents “another approach” to the study of valuation. This conclusion is bolstered by the fact that much of the language used is different and that we use different tools such as decision trees and the binomial lattice. The truth of the matter, however, is that we are still doing DCF analysis. The innovation in this chapter is that we use observed market prices for derivative securities to *improve* our DCF analysis (not replace it).
- If it has been a while since you studied derivatives or if you have never done so, you will want to spend some extra time on Appendix A, where we provide a basic review of options and their payoffs. You may also find it helpful to visit one of the many excellent Web-based resources for a review of derivative contracts.⁴
- Finally, in this chapter, we use the binomial lattice for the first time. This will be a new tool for many readers. Consequently, you would do well to spend the time you need to fully understand this tool, because it will also be used in the next two chapters.⁵ The importance to the real options approach of understanding the use of this tool is analogous to the importance of understanding how to do time value of money calculations in DCF analysis, so “don’t leave this chapter without it”!

This chapter is organized as follows: Section 2 discusses the certainty-equivalence method for valuing risky future cash flows. This is one of the three discounted cash flow models discussed in earlier chapters and is commonly used in situations where derivative markets exist. Section 3 provides an example where we value a risky investment whose primary source of risk is that associated with the price of a commodity. As we show, the presence of forward market prices to hedge this risk greatly reduces the complexity of the task (if you are not familiar with the concept of financial hedging with forward and futures contracts or if you would benefit from a review, see the Technical Insight box

⁴ For example, see <http://www.financialpolicy.org/dscprimer.htm>.

⁵ If you find you need some extra help with this, see Chapter 10 in Robert L. McDonald, *Derivative Markets*, 2nd ed. (Boston: Addison Wesley, 2006).

TECHNICAL INSIGHT

A Primer on Forward and Futures Contracts

Forward contracts and futures contracts are similar in that they both represent agreements to buy or sell an asset at a specified time in the future for a specified price. The principal difference between a futures contract and a forward contract is that the *futures contract* is traded on an organized exchange and the forward is not. Instead, the *forward contract* represents a private or over-the-counter agreement between two financial institutions or between a financial institution and one of its corporate clients.

In addition to market differences, forward contracts usually have only one specified date for delivery of the commodity on which the contract is written, whereas a futures contract may offer a range of delivery dates. Still another important difference between forwards and futures is the method of settlement for the difference between the value of the underlying commodity and the agreed-upon future delivery price. For a forward contract, this difference is settled at the termination of the contract; in the case of a futures contract, it is settled *daily* through a procedure called *marking-to-market*. Finally, futures contracts, unlike forwards, are usually terminated or closed out prior to delivery; the physical commodity is not (usually) delivered.

The following exhibit summarizes the unique features of forward and futures contracts:

	Forward	Futures
Market	Private contract between two parties	Traded on an organized exchange
Contract	Not standardized	Standardized
Delivery	Generally on one delivery date	Range of delivery dates
Settlement	At the end of the contract	Daily (marked-to-market)
Termination	Delivery of commodity or final cash settlement	Generally, the contract is closed out prior to maturity

You might think that the market for immediate or current delivery of a commodity would be called the “current market,” in contrast with the forward or futures market. However, this is not the case. This market is referred to as the *spot market*.

Each forward or futures transaction has a buyer and a seller. The party that agrees to buy the commodity at a future date is said to have a *long position*, whereas the party that agrees to sell the commodity in the future is said to have taken a *short position*.

entitled “Hedging Financial Risks”). Section 4 opens our discussion of options by providing an example of the valuation of an equity investment in a levered oil field. This example illustrates how firms can use options to capture the value of decision flexibility inherent in many investment opportunities. In the context of this example, we introduce the binomial option valuation technique, which is widely used to value financial options as

well as real investments that have option-like components (Appendices B and C provide details for anyone who would like to understand the binomial model in more depth). Section 5 discusses a number of caveats or limitations of the methods we propose in this chapter. Specifically, it is generally impossible to find market prices of traded commodities that exactly match the attributes of the investments we are trying to value. As a result, the methodology has sources of error that are easily identified. We should be reminded, however, that the traditional discounted cash flow evaluation is also subject to forecast errors when cash flows projections are made. In Section 6, we demonstrate the use of the binomial option pricing model as a tool for valuing options. Finally, Section 7 provides summary comments.

2 THE CERTAINTY-EQUIVALENCE METHOD

The approach that we will use to value investments in this chapter is a special case of the **certainty-equivalence (CE)** approach that requires that we first *risk-adjust* the risky future cash flows, creating what we call *certainty-equivalent* cash flows. A **certainty-equivalent** cash flow is defined as the certain cash flow that has the same value to the recipient as the uncertain cash flow that is being evaluated. For example, suppose the holder of a lottery ticket that is worth either \$70 or \$100 with equal probability is willing to accept a sure payment of \$82 in exchange for the ticket. We can say that the certainty equivalence of the uncertain cash flow to the lottery ticket is \$82, which is lower than the \$85 expected payoff (i.e., $\$85 = .5 \times \$100 + .5 \times \$70$). The holder of the lottery ticket is willing to take less than the expected value because of his aversion to risk.

The certainty-equivalence approach to valuation is fundamentally the same as the traditional discounted cash flow (DCF) approach except that the former adjusts the cash flows for risk rather than adjusting the discount rate. In most cases, either adjustment is appropriate, but in general you should *not* risk-adjust *both* the cash flows *and* the discount rate. As it turns out, for the natural resource valuation problems that we consider in this chapter, it is

TECHNICAL INSIGHT

Hedging Financial Risks

Hedging is the process of reducing the firm's risk exposure by transferring part or all of it to another firm or entity. For some risks, such as the risk of fire or property damage to the firm's facilities, the hedge might entail the purchase of insurance. Other risks can be hedged in the financial derivatives market.

For example, a firm that sells its products and services to international clients who pay for them at a later date in a foreign currency is exposed to currency risk. Similarly, firms that need to borrow money in the future may be exposed to interest

rate risk. Both the risk of adverse changes in foreign exchange rates and a rise in interest rates can be hedged using financial markets.

Consider how the hedging process would work using *forward contracts*, which represent agreements to buy or sell an asset at a certain time in the future for a certain price. To hedge, we combine a future commitment that the firm has to sell (buy) an asset with the acquisition (sale) of a forward contract. By locking in the price today that will be received (paid) in the future, the firm effectively eliminates its price risk.

more convenient to risk-adjust the cash flows rather than the discount rate, because forward prices can be used to help us make the risk adjustment.

Forward Prices as Certainty-Equivalent Cash Flows

The certainty-equivalence concept is quite useful in those situations in which the certainty-equivalent cash flow of an investment can be observed and calculated using observed market prices. Indeed, specifying the certainty equivalence of a future cash flow is preferred to specifying the expected cash flow, because the valuation of certainty-equivalent cash flows involves discounting at the risk-free rate and thus does not require an estimate of the appropriate risk-adjusted discount rate.

In some cases, managers can estimate the certainty equivalence of a cash flow more accurately than they can its expected value. This will generally be the case when the risk associated with a cash flow is closely related to the risk of securities traded in the derivative markets. To illustrate this, let's first consider the **forward prices** of commodities like oil, grains, or metals. The forward price of such a commodity is the price, which is set today and at which market participants are willing to buy and sell that commodity for delivery at some future date. In other words, it is the certain price today, which has the same current value as the uncertain price that will occur in the future. As such, forward prices can be viewed as the certainty equivalents of the uncertain prices that will be realized in the future.

To better understand this concept, consider an investment that generates 100,000 barrels of oil in one year. Because oil prices are uncertain, the cash flows from this investment are uncertain. However, suppose that there exists a forward market that allows you to buy or sell oil, for delivery in one year, for a price of \$60/barrel. In this case, we would say that the certainty equivalence of the uncertain future oil price in one year is \$60 and that the certainty equivalence of the investment's cash flow is \$6 million ($\$60/\text{barrel} \times 100,000$ barrels). The owner of the oil can obtain this amount for certain by selling his or her oil in the forward market rather than facing the uncertain cash flow associated with the uncertain price of oil.

It should be stressed that the forward price is generally not equal to the expected price in the future. For example, suppose the price of oil next year is equally likely to be either \$75/barrel or \$55/barrel, implying that the expected price of oil is \$65/barrel. If market participants have no aversion to risk, then this is also likely to be the forward price. However, if a sufficient number of risk-averse oil producers choose to hedge their oil price risk by selling forward contracts, the forward price will be below

Did you know?

Where Is the World's Largest Commodity Futures Market?

The New York Mercantile Exchange Inc., or NYMEX, is the world's largest physical commodity futures exchange and a major trading forum for energy and precious metals. The exchange pioneered the development of energy futures and options contracts and offers options on all major futures contracts: light, sweet crude oil; Brent crude oil; heating oil; unleaded gasoline; natural gas; coal; gold; silver; platinum; copper; and aluminum. Also available are two crack spread options contracts—one for the differential, or the spread, between heating oil futures and light, sweet crude oil futures and one on the New York Harbor unleaded gasoline/light, sweet crude spread.

Did you know?

Southwest Airlines' Oil Price Hedge

Southwest Airlines's successful efforts to hedge the risk of rising jet fuel provide a classic example of the use of forward contracts. Specifically, the company used forward contracts to lock in the price of jet fuel for much of its expected fuel needs for 2005–2009. Southwest locked in 65% of its expected jet fuel needs in 2006 at \$32 a barrel, more than 45% of its needs for 2007 at a maximum price of \$31 per barrel, 30% of its planned 2008 fuel purchases at \$33 per barrel, and 25% of its 2009 fuel needs at \$35 per barrel.

(Source: http://money.cnn.com/2005/04/14/news/fortune500/southwest_oil/)

\$65/barrel. The actual price that is realized in the forward market will be determined by supply-and-demand conditions that relate to the number of producers who want to hedge their oil price exposure by selling oil in the forward market and the number of oil users (like airlines) and speculators (like hedge funds) that are willing to buy oil in the forward market. Although the equilibrium forward prices can, in theory, equal the expected price, in general this will not be the case.

3 USING FORWARD PRICES TO VALUE INVESTMENT PROJECTS

Natural resource investments, such as oil and gas fields, illustrate how forward prices are used in discounted cash flow valuation. Application of this method is made possible because oil and gas are commodities that have well-developed forward and futures markets; consequently, managers in these industries can use prices from these markets to calculate certainty-equivalent cash flows.

Before valuing an investment with the certainty-equivalence approach, it is useful to first review how analysts value oil and gas investments using the traditional DCF approach. With the traditional method, expected cash flows are calculated by multiplying expected prices by expected quantities and then discounting these cash flows using risk-adjusted discount rates. This traditional DCF method requires the analyst to make a number of estimates, including estimates of the underlying determinants of both the investment's expected cash flows (based on forecasted prices in combination with estimated extraction rates and operating costs) as well as the appropriate cost of capital to use in discounting the expected cash flows. In contrast, by utilizing information from the oil futures market, we can apply the certainty-equivalence method described in the previous section without having to forecast future oil prices or the appropriate discount rate, because all this information is impounded in forward prices. In the oil and gas industry, this practice is referred to as "pricing against the forward price curve." The following example illustrates how this can be done.

Example

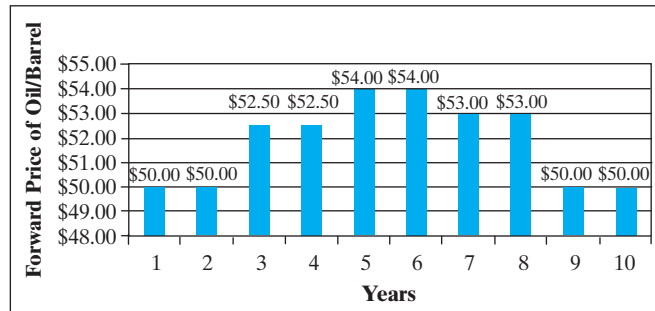
Cutter Exploration and Drilling Company is considering a promising opportunity to develop acreage in the Austin chalk region of central Texas. This area has been in production for many years but has been plagued by recovery problems. Cutter has developed a new procedure for use in recovering oil from this type of geologic structure that has proven to be very successful in similar situations along the Louisiana Gulf Coast. The procedure is applicable for extracting oil from an oil lease Cutter is considering near Giddings, Texas. Let's see how we might use futures prices and the certainty-equivalence model to evaluate the investment.

Cutter's geologists are confident that the Giddings lease has 1,000,000 barrels of recoverable oil reserves, which can be extracted at a rate of 100,000 barrels per year over a period of 10 years using Cutter's new technology. To acquire this

Did you know?

What Fraction of an Underground Oil Reserve Is Recoverable?

Total oil reserves and recoverable reserves are not the same thing. For example, using standard recovery techniques, it is possible to extract no more than 15% to 20% of the total reserves found in the Austin chalk region near Giddings, Texas. However, using newly developed horizontal drilling and fracturing methods, the extraction percentage increases to about 50%.

Figure 1 Forward Price Curve for Crude Oil

opportunity, Cutter will have to pay \$7,000,000 for the lease (i.e., the right to drill for and produce the oil) and will incur drilling costs of \$12,000,000. For simplicity, we assume that the total investment of \$19,000,000 is spent at the beginning of the first year of the investment (i.e., at time $t = 0$) and that the cost of extracting and transporting the oil is \$28 per barrel. Note that we assume that the quantity of oil to be extracted is known and that the costs of extracting it are fixed at \$28 per barrel. These assumptions leave only the price of oil as a source of risk to the investment. We return to this point later in the chapter.

Cutter's banker is willing to write a forward contract for the delivery of 100,000 barrels per year over the next 10 years. The "forward price curve" specified in this contract is found in Figure 1. For example, oil produced in Years 1 and 2 has a forward price of \$50 per barrel, oil sold in Years 3 and 4 is priced at \$52.50 per barrel, and so forth. Currently, the risk-free rate of interest on 10-year U.S. government bonds is 5%.

We can use the certainty-equivalence method to estimate the value of the investment to Cutter using the three-step process outlined in the Chapter Overview:

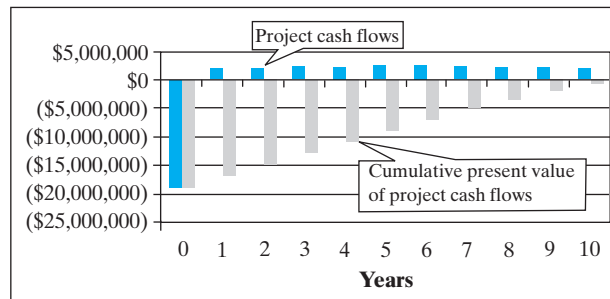
- Step 1:** Forecast the amount and timing of future cash flows. The most difficult part of this step has already been done, in that the total production volume is assumed to be 100,000 barrels per year over the next 10 years.
- Step 2:** Use the forward price curve to calculate the certainty-equivalent future cash flows from the project.
- Step 3:** Discount the certainty-equivalent cash flows back to the present using the risk-free rate of interest.

Figure 2 depicts the cash flows generated by our simple investment example. Because the forward price for oil is equal to the certainty-equivalent price, the annual certainty-equivalent cash flow in Years 1 and 2 equals $\$50 - \$28 = \$22$ per barrel times 100,000 barrels, or \$2,200,000. In Years 3 through 6, the annual cash flow rises because the forward price increases. By discounting these certainty-equivalent cash flows for Years 1 through 10 using the risk-free rate, we get a value for the project of \$18,437,605. Subtracting the \$12,000,000

Did you know?

What's the "Lingo" of the Forward Price Curve?

Forward price curves can be either upward sloping (i.e., forward prices for more distant periods are higher) or downward sloping. An upward-sloping forward curve is said to be in **contango**, and a downward-sloping curve is said to be in **backwardation**.

Figure 2**Natural Resource Investment Project Analysis**

cost of drilling and completing the well and the \$7,000,000 cost of acquiring the lease from the \$18,437,605 estimated value of the production during years 1 through 10 generates a negative net present value for the property of (\$562,395), found in Table 1. Because the NPV is negative, the analysis indicates that Cutter should reject the investment. In the next section, we investigate the validity of this analysis more fully.

In the preceding example and for the balance of this chapter, we will ignore the effects of taxes and the use of debt financing in order to simplify the analysis. For a discussion of the certainty-equivalence method including debt tax shields, see the Technical Insight box two pages ahead.

Convincing Your Skeptical Boss

Now suppose that you completed your analysis of Cutter's investment opportunity and delivered it to your boss. To your surprise, he strongly disagrees with your analysis. He believes that oil prices will average close to \$60 per barrel over the next 10 years and also believes that the forward prices you have used to forecast cash flows are not relevant to the project valuation because he does not plan to hedge the price of oil (that is, he does not plan to take up the bank's offer and commit to selling the production at the quoted forward prices). Based on his more optimistic forecast of \$60 per barrel and a discount rate of 10%, he estimates that the project has a positive NPV of \$662,615.⁶

How should you convince him that his analysis is wrong and your analysis is correct? As we will show, since the derivatives valuation approach uses market prices, the analysis implies that the derivatives market offers a better investment opportunity regardless of your boss's beliefs about future oil prices. By showing your skeptical boss that there exists a superior alternative investment opportunity, we should be able to convince him that the investment under evaluation is not attractive.⁷ Specifically, we will devise an alternative investment strategy involving the purchase of a risk-free

⁶ The annual cash flow is calculated as follows: 100,000 barrels (\$60/barrel – \$28/barrel) = \$3,200,000. Discounting these cash flows using the 10% cost of capital estimate provides a present value of \$19,662,615. Subtracting the total investment costs of \$19 million produces an NPV of \$662,615.

⁷ Keep in mind, however, that by showing your boss that he is in error, you may be proven right, but may also find yourself working from a new office in one of the company's most obscure locations!

Table 1 Project Cash Flows and Present Value

Year	Forward Price/Barrel ^a	Project Free Cash Flows ^b	Present Values ^c	Cumulative Present Values
0		\$ (19,000,000)	\$ (19,000,000)	\$ (19,000,000)
1	\$ 50.00	2,200,000	2,095,238	(16,904,762)
2	50.00	2,200,000	1,995,465	(14,909,297)
3	52.50	2,450,000	2,116,402	(12,792,895)
4	52.50	2,450,000	2,015,621	(10,777,274)
5	54.00	2,600,000	2,037,168	(8,740,106)
6	54.00	2,600,000	1,940,160	(6,799,946)
7	53.00	2,500,000	1,776,703	(5,023,242)
8	53.00	2,500,000	1,692,098	(3,331,144)
9	50.00	2,200,000	1,418,140	(1,913,004)
10	50.00	2,200,000	1,350,609	(562,395) ^d

^aThe prices in the forward price curve have been set through a contract with the bank.

^bFor Year 0, the project cash flow consists entirely of the \$12 million cost of drilling plus the \$7 million cost of acquiring the drilling lease (rights). In Year 1, the cash flow equals the difference in total revenues (\$5 million = \$50/barrel $\times$ 100,000 barrels), and the cost of extracting and transporting the oil (\$28/barrel $\times$ 100,000 barrels = \$2.8 million), or \$2.2 million. Note that we assume a zero tax rate.

^cBecause the oil prices are hedged, they don't involve any future risk. Thus, we discount future cash flows using the risk-free rate of interest.

^dBecause these are cumulative present values, the value for Year 10 incorporates the present values of all 10 years of cash flows, including the initial outlay such that this is the NPV. The present value of the cash flows for Years 1 through 10 is \$18,437,605. Subtracting the initial outlay for the project from (\$19,000,000), we get the NPV of (\$562,395).

bond and a series of long positions in oil forward contracts that exactly replicates the cash flows of the oil investment project over the next 10 years. As we will show, this investment costs less than the \$19 million required to make the oil investment, but generates identical cash flows.

Here's how you can construct the alternative investment:

- First, the company purchases a portfolio of risk-free bonds that generate annual cash flow equal to the project's certainty-equivalent cash flows. For example, you could purchase 10 zero coupon bonds with maturity values that are equal to the 10 annual cash flows found in Table 1. This portfolio would cost an amount equal to the present value of the 10 annual cash flows, or \$18,437,605.
- Next, in addition to buying the portfolio of bonds, the company also enters into a series of forward contracts with the bank, in which the bank agrees to sell 100,000 barrels of oil to be delivered at the end of each of the next 10 years at the prices found in the forward price curve in Figure 1. The profit (or loss) on these contracts will be the difference between the uncertain future market price that is realized at those delivery dates and the forward price per barrel specified in the contract.

TECHNICAL INSIGHT

The Certainty-Equivalence Approach with Debt and Taxes

Accounting for an investment's debt tax shield is straightforward with the certainty-equivalence approach. If the free cash flows of an investment are calculated as certain equivalents, they should be discounted using the risk-free rate for the cost of both equity and debt. Adjusting for the tax deductibility of interest costs results in an after-tax WACC equal to the following:

$$k_{WACC} = r_f - \left(\frac{\text{Tax}}{\text{Rate}} \right) r_f \left(\frac{\text{Debt}}{\text{Value}} \right) = r_f \left[1 - \left(\frac{\text{Tax}}{\text{Rate}} \right) \left(\frac{\text{Debt}}{\text{Value}} \right) \right]$$

where r_f is the risk-free rate of interest. For example, if the tax rate is 20%, the risk-free rate is 5%, and the debt-to-value ratio is 40%, then the WACC that would be used with the certainty-equivalence approach is calculated as follows: $5\%(1 - .2 \times .4) = 4.2\%$.

We can also calculate the debt tax shield using the APV approach, which separately values the unlevered or operating cash flows and the debt tax shield. When using this approach, we recommend that the first component be valued by discounting the before-tax certainty-equivalent free cash flow at the risk-free rate of interest. We do not, however, recommend calculating the certainty equivalence of the interest tax savings component; this should be calculated and discounted at the cost of debt.

To see how this works, let's assume your boss is right, and the price per barrel of oil climbs to \$60; thus, the oil field generates a cash flow for Year 1 equal to \$3,200,000.

$$\text{Cash Flow}_1 = (\$60/\text{Barrel} - \$28/\text{Barrel}) \times 100,000 \text{ Barrels} = \$3,200,000$$

Alternatively, if we had invested in the bond and the forward contracts, our bond payment would be \$2,200,000 (the Year 1 cash flow based on the forward price of \$50/barrel) and the profit from our forward contract would be $(\$60 - \$50)/\text{barrel} \times 100,000 \text{ barrels} = \$1,000,000$. Added together, the Year 1 cash flow from the bond investment and the forward contract is \$3,200,000, which is identical to the cash flow from the oil investment. Panel a of Table 2 contains the calculations for both the replicating strategy (i.e., bond plus forward contracts) and the actual project, where the price of oil is expected to equal \$60/barrel every year.

Now consider a different scenario for Year 1: Let's assume that OPEC falls apart and crude oil prices drop to only \$35/barrel. In this scenario, owning the well generates cash flow of only \$700,000, i.e.:

$$\text{Cash Flow}_1 = (\$35/\text{Barrel} - \$28/\text{Barrel}) \times 100,000 \text{ Barrels} = \$700,000$$

From our alternative investment, we would earn \$2,200,000 from the bond payment and lose $(\$1,500,000) = (\$35/\text{barrel} - \$50/\text{barrel}) \times 100,000 \text{ barrels}$ from our forward contracts, netting the identical \$700,000 cash flow. Panel b of Table 2 contains the cash flows for both the replicating strategy (using bonds and forward contracts) and the actual project, under the assumption that the price of crude equals \$35/barrel in each of the next 10 years.

Table 2 Replicating Project Cash Flows Using Forward Contracts

Panel a. Expected Price of Oil Equal to \$60/Barrel				
	(A)	(B)	(A + B)	
Year	Bond Payouts	Gain/Loss on Forwards	Replicating Cash Flows	Actual Project Cash Flows
1	\$2,200,000	\$ 1,000,000	\$3,200,000	\$3,200,000
2	2,200,000	1,000,000	3,200,000	3,200,000
3	2,450,000	750,000	3,200,000	3,200,000
4	2,450,000	750,000	3,200,000	3,200,000
5	2,600,000	600,000	3,200,000	3,200,000
6	2,600,000	600,000	3,200,000	3,200,000
7	2,500,000	700,000	3,200,000	3,200,000
8	2,500,000	700,000	3,200,000	3,200,000
9	2,200,000	1,000,000	3,200,000	3,200,000
10	2,200,000	1,000,000	3,200,000	3,200,000
Panel b. Expected Price of Oil Equal to \$35/Barrel				
	(A)	(B)	(A + B)	
Year	Bond Payouts	Gain/Loss on Forwards	Replicating Cash Flows	Actual Project Cash Flows
1	\$2,200,000	\$ (1,500,000)	\$700,000	\$700,000
2	2,200,000	(1,500,000)	700,000	700,000
3	2,450,000	(1,750,000)	700,000	700,000
4	2,450,000	(1,750,000)	700,000	700,000
5	2,600,000	(1,900,000)	700,000	700,000
6	2,600,000	(1,900,000)	700,000	700,000
7	2,500,000	(1,800,000)	700,000	700,000
8	2,500,000	(1,800,000)	700,000	700,000
9	2,200,000	(1,500,000)	700,000	700,000
10	2,200,000	(1,500,000)	700,000	700,000

What this simple example illustrates is that we *can* replicate the cash flow stream from the risky oil drilling investment using risk-free bonds and forward contracts to purchase oil. Note the key difference between the two alternatives, however: The up-front cost of the replicating strategy using the forward market to purchase oil and investing in a risk-free bond is equal to the cost of the bond, or \$18,437,605, because the forward contracts have no cost associated with them at the time they are initiated. On the other hand, it costs \$19,000,000 to purchase and develop the oil field investment. Clearly, the

replicating strategy dominates investing in the Giddings project, as it generates *identical* annual cash flows, yet the replicating strategy costs \$562,395 less (which is the negative-NPV-loss associated with making the investment).

4 USING OPTION PRICES TO VALUE INVESTMENT OPPORTUNITIES

The investment considered in the previous section was quite simple. The buyer did not use any debt financing, and the implementation of the investment required no managerial choices beyond the initial decision to undertake the investment. The key insight we gained from our analysis was that we could value the investment using market prices from the forward price curve for oil.

This section illustrates how a similar process can be used to value investment opportunities that have option-like components. Specifically, we show how companies can use **financial options** (i.e., options that are traded in financial markets) to replicate investment cash flows and to value investments in much the same way in which we used forward contracts in the earlier example.

We will start our discussion by considering the **default option** embedded in debt contracts. This is an example of what we refer to as a **contractual option**, which is an option that arises because of the design of the contract that defines the terms of an investment. To illustrate the importance of the default option, we will consider an investment that is financed with *nonrecourse debt*, or debt that is collateralized only by the specific assets being financed, with no separate guarantee by the firm. When a firm uses nonrecourse debt to finance an investment, it has the option to default on the debt and simply walk away from the investment if it is doing especially poorly. As we will show,

TECHNICAL INSIGHT

Option Terminology

The type of option contract—Put or call. A *call (put)* option gives its owner the right, but not the obligation, to buy (sell) a specified asset at a contractually agreed-upon price (commonly referred to as the option's *exercise price*) within a specified period of time. Appendix A provides a brief review of option-contract payoffs.

The underlying asset. The *underlying asset* is the common stock, tract of land, etc., that the owner of the option contract buys or sells if the contract is exercised. We say the option is *written* on the underlying asset.

The expiration date of the option. Options have a finite life during which they can be exercised. The final date on which the option

can be exercised is referred to as its *expiration date*.

The exercise (or strike) price. The *exercise (strike) price* is the price at which the option owner may buy or sell the underlying asset on which the option is written. We will use the symbol K to refer to the option's exercise price.

The terms of exercise—American or European. A *European option* provides the holder with the right to exercise the option only on its expiration date. The *American option* allows the holder to exercise the option at any time up through the expiration date.

For an in-depth review of option contracts, refer to Appendix A.

observed option prices can sometimes be used to determine the value of the firm's equity investment in such a project.

Option Value and Nonrecourse Financing

To illustrate the importance of incorporating information from the prices of financial options in the evaluation of productive investments, consider the following investment opportunity:

- For a price of \$5 million, you have the opportunity to purchase the equity of the Cotton Valley oil field, which contains 5 million barrels of reserves. Note that the oil field investment will be partially financed by borrowing, so this is a levered investment.
- The oil can be extracted in one year at a cost of \$12 per barrel, for a total cost of \$60 million = \$12/barrel $\times$ 5 million barrels. For simplicity, we assume that the extraction costs are paid at the end of the year and at the time the oil is produced.
- The current owner borrowed \$200 million to finance the purchase of the field. If you purchase the equity, you will assume this debt, which requires a onetime payment at the end of the year of \$220 million (including interest of 10%).
- The debt is *nonrecourse*, which means that the debt holders cannot lay claim to the acquiring firm's other assets if it defaults on the debt payments. They can lay claim only to the oil field. As an equity investor, your firm has the option not to repay the loan and to walk away from the investment if it is in your financial interest to do so. If you chose to do so, you would forfeit any rights of ownership to the oil field, which would transfer to the lender. For example, this would be the case if the revenues generated by the oil field at the end of the year turn out to be worth less than \$280 million—the \$220 million needed to retire the note plus the \$60 million needed to develop the property and produce the oil. Note that the \$280 million “breakeven” point equals \$56/barrel = \$280 million/5 million barrels.
- The forward price of oil to be delivered in one year is \$54.17, and the price of a call option to purchase one barrel of crude oil in one year, at an exercise price of \$56/barrel, is \$1.75.

Valuing the Investment Using Forward Prices

Before proceeding with our formal analysis of the default option, we begin by explaining why the approach from Section 3 that used forward prices is not appropriate in this case. In the previous example, we valued the investment by assuming that the oil price risk is hedged or sold in advance in the forward market. However, in this particular case, if the oil from the field is sold today for the forward price of \$54.17, the equity investment will have no value. To understand this, note that if the investor sells the crude oil forward for \$54.17/barrel, this will guarantee revenues of \$270,850,000 = 5,000,000 barrels $\times$ \$54.17/barrel. The equity cash flow at the end of one year for the hedged investment is then:

$$\begin{aligned}\text{Equity Cash Flow} &= \text{Revenues} - \text{Extraction Costs} - \text{Note Payoff} \\ &= \$270,850,000 - 60,000,000 - 220,000,000 = (\$9,150,000)\end{aligned}$$

Hence, if the oil is sold forward, the equity investment generates a negative cash flow of (\$9,150,000) if the investor chooses not to default. As a result, the equity investor will in fact default on his debt obligation, so the actual cash flow to the equity holders is zero.

Because of the option to default, the investment in the project's equity has a higher value if the oil price risk is not hedged. Indeed, when we value this investment opportunity in a way that incorporates the option to default on the debt, we will see that the equity investment does have a positive value when the oil price risk is not hedged.

Analyzing Equity Value with the Default Option—Managing the Project

We begin our analysis by showing that the payoff to the equity in the project is identical to an investment in a one-year call option on 5 million barrels of oil with an exercise price of \$56/barrel (i.e., the cost of extracting a barrel of oil from the oil field in one year).

- If the price of oil exceeds \$56/barrel, the option will be exercised and will generate a profit equal to the price of oil times 5 million barrels minus \$280 million (the \$56 exercise price per barrel times 5 million barrels). This profit is identical to the profit from the equity investment, which also generates cash flows equal to the price of oil times 5 million minus \$280 million (the sum of the \$60 million in extraction costs plus the \$220 million needed to retire the note).
- If oil prices fall below \$56/barrel, the option expires worthless. When this is the case, the equity investor defaults on the debt and walks away from the investment.

Just as having forward prices simplified the analysis of Cutter Exploration Company in the example discussed earlier, having observable option prices simplifies the valuation of this equity investment. In this case, the **tracking portfolio**—that is, the portfolio of traded securities that replicate the real investment cash flows—for the oil investment consists of a portfolio of call options on 5 million barrels of crude oil with an exercise price of \$56/barrel. If these call options have a price of \$1.75/barrel, then the tracking portfolio is worth \$8.75 million = \$1.75 per barrel $\times$ 5 million barrels of oil. Because the equity investment in the project has the same cash flows as the tracking portfolio, it too is worth \$8.75 million. Given that the equity of the Cotton Valley oil field investment can be acquired for \$5 million, this seems like a pretty good investment. Indeed, the NPV of the investment is \$3.75 million = \$8.75 million – \$5 million.

Convincing Your Skeptical Boss

Again, because this kind of analysis is relatively new, we might expect to encounter a skeptical boss, reluctant to accept the valuation of the equity in the project using the option pricing approach. Because this is a positive-NPV investment, we cannot create a better transaction using derivatives, as we did earlier in the Cutter example by using risk-free bonds and forward contracts. Instead, to convince your skeptical boss to take the investment, we should show her how to use the derivative market to hedge the risk associated with this investment and thereby guarantee that the NPV of the investment is realized.

To lock in the positive NPV associated with this investment, we acquire the equity position in the oil property for \$5 million and at the same time write (sell) call options on 5 million barrels of oil for \$1.75 per barrel. Because the option transaction generates a total cash inflow of \$8.75 million = 5 million $\times$ \$1.75, the combination of these two transactions nets \$3.75 million in the current period. As we illustrate in Table 3, despite this initial cash inflow, the combination of these two investments generates no cash outflows in the following year, because the cash flows of the two investments exactly offset each other.

TECHNICAL INSIGHT

Settlement Differences in Forward Versus Option Contracts and Discounting

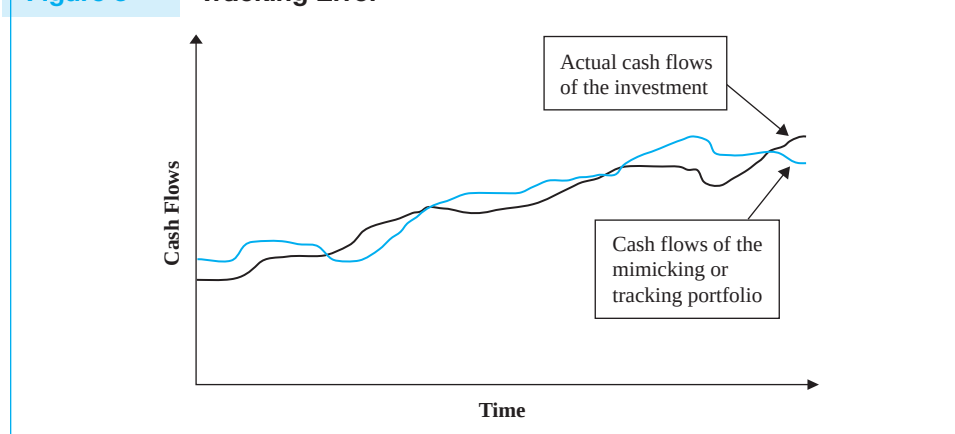
The cash flows associated with forward and option contracts change hands at different times. When a forward contract is initiated, no cash changes hands until the contract delivery date. Consequently, when forward prices are used to calculate the certainty equivalent of future project cash flows, we discount the resulting cash flows back to the present using the risk-free rate to determine their value. However, when the market prices of options are used to value an investment, we do not discount them. The reason for this difference is that with options, the buyer of an option pays up front—i.e., the option price we observe is the discounted value of the certainty-equivalent payoff of the option.

Table 3 Locking in a Positive NPV Using Options—Cutter Example

Year	Hedging Strategy	Cash Flows
0	(1) Sell call options on 5 million barrels of crude oil for \$1.75 each	\$8.75 million
	(2) Purchase the oil investment	\$ (5 million)
	(3) Net cash flow for Year 0	\$3.75 million
Cash Flows for Alternative Crude Oil Prices		
		\$60/barrel
1		\$ 56/barrel
	(1) Payoff (loss) from sale of call options	$(\$4) \times 5 \text{ million} = (\$20 \text{ million})$
	(2) Payoff (loss) from the oil investment	$\$300 \text{ million} - \$280 \text{ million} = \$20 \text{ million}$
	(3) Net cash flow for Year 1	\$0.00

5 CAVEATS AND LIMITATIONS—TRACKING ERRORS

Most of our discussion up to this point has assumed that it is possible to construct a portfolio of traded financial securities whose cash flows are a *perfect match* for the cash flows of the real investment we are trying to value. In practice, it is not always possible to match cash flows perfectly with the payoffs of traded financial securities. As a consequence, the value of the tracking portfolio will differ from that of the investment. Figure 3 illustrates the problem of tracking error. The black line depicts the true value of the project over time, and the blue line reflects the value of the tracking portfolio over a period of time. Obviously, the greater the difference between the two lines, the less precise the valuation of the investment using the value of the tracking portfolio.

Figure 3**Tracking Error**

There are three fundamental reasons why the cash flows of the tracking portfolio will deviate from those of the investment being analyzed. The first relates to the reliability of market prices for derivatives. The second relates to what we will refer to as *omitted project risks*, which are simply sources of risk in the investment's cash flows that are not captured in the cash flows of the tracking portfolio. Finally, there is something that derivatives traders refer to as *basis risk*, which arises out of differences in the specific nature of the assets underlying the derivative contracts and the assets that underlie the real investment being evaluated. We describe each of these sources of tracking error in some detail below.

How Liquid Are Futures, Forward, and Option Markets?

If derivative markets are very illiquid, the market prices may not provide a reliable estimate of the certainty-equivalent prices. Our analysis assumes that the investor can, in fact, transact at the quoted financial market prices. For example, suppose we are considering a project to extract two billion barrels of oil over a five-year period (a bit more than one million barrels per day). Although the derivative markets have sufficient depth to hedge multibillion-dollar oil price exposures, it may not be possible to hedge such a huge exposure without significantly affecting forward prices. Because of this, large oil producers tend to hedge very little, and, as a result, they do not believe the derivatives approach is applicable for their investments and prefer to use the traditional DCF approach. Although they may use futures prices as a guide to their estimates of expected oil prices, they also use their own judgment about expected future oil prices.

Uncertain Quantities and Operating Costs

Our second caveat has to do with project risks that have been omitted from our analysis. Here, tracking errors arise from sources of variation in cash flows that cannot be known in advance and consequently cannot be used to determine the tracking portfolio. Specifically, although the tracking portfolio procedure can be extremely valuable in dealing with the commodity price risks in capital investments, typically there are other sources of risk that affect an investment's cash flows that are not driven by commodity prices. Two important

sources of noncommodity price risk are the quantity of the commodity produced and the operating costs associated with extracting and processing the commodity.

In our oil and gas examples, you may have noted that we assumed that the quantity of oil and the cost of extraction were both known with certainty, so that the only source of risk was the uncertain price of crude oil. In practice, valuing an oil field (or any investment in a commodity industry) requires estimates of the quantity of oil produced and the costs of extracting the oil. To accommodate the effects of uncertain production quantities, managers may opt for conservative estimates of those quantities to adjust for this source of uncertainty.⁸ One might also use relatively conservative estimates of extraction costs (i.e., costs on the high end of what is expected) for the same reason.

Basis Risk

The final source of tracking error that we consider is something analysts refer to as **basis risk**, which, as we noted earlier, arises out of differences in the specific nature of the assets underlying the derivative contracts and the assets that underlie the real investment being evaluated. We will discuss three sources of such differences: product quality, geographic location, and contract terms.

Differences in Product Quality

Although the list of traded derivatives on commodities is quite lengthy, it is not encyclopedic. This means that the analyst will often have to use a surrogate commodity contract, because a derivative contract for the exact commodity that underlies the risks of the investment being analyzed may not exist. For example, if the investment involves the production of heavy crude oil, the analyst might use derivative contracts based on light, sweet crude, because this is the contract that is traded on the New York Mercantile Exchange (NYMEX). The problem is that the price of light, sweet crude is imperfectly correlated with the price of other forms of crude oil. Consequently, the resulting valuation of the investment using the light, sweet crude contracts will differ from the true value of the investment because of the differences in the *quality of the products* underlying the investment and the derivative contract.

Differences in Geographic Locations

Basis risk also arises out of *geographic differences*. For example, crude oil futures are priced for delivery in Cushing, Oklahoma, and natural gas futures are priced for delivery at Henry Hub in Louisiana. Obviously, differences in the physical location of crude oil or natural gas from the standard locations used in the option contracts give rise to tracking errors. In this instance, the tracking error is due to delivery costs.

Although the problem of tracking errors due to geography can be resolved by considering delivery costs, the problem is not inconsequential. Delivery costs vary with supply-and-demand conditions, and in some extreme instances, the costs can become prohibitive. For example, during the California energy crisis in 2000, the costs of delivering electric power to the residents of the state rose to astronomical levels as the power grid became overloaded.

⁸ However, one would not want to use quantity estimates that are substantially lower than expected quantities, because the quantity of oil that is produced is not likely to be correlated with the aggregate economy. In other words, quantity is likely to be a low or zero beta risk.

Differences in Contract Terms

These are cases where the match between traded options and the option component of the investment that is being valued is far from perfect. For example, suppose we can find prices of call options on oil with an exercise price of \$50, but there are no traded options with an exercise price of \$56, which is the price we need to solve our valuation problem. An even more common problem is that the analyst is evaluating an investment that offers cash flows distributed over many years, whereas traded options have maturities extending out only one year. In such cases, we will need an option pricing model to value the investment, which is the topic we discuss in the next section.

The Relation Between Tracking Error and Imperfect Comps

The astute student of finance will recognize that the problem of finding an appropriate tracking portfolio is analogous to the problem of finding appropriate comparison firms that can be used to value a business. As we mentioned in earlier chapters, it is generally impossible to find comparable companies that provide a perfect match when we are trying to identify either the appropriate valuation multiple or the appropriate risk-adjusted cost of capital. In general, managers consider a variety of comps and then use their judgment to come up with a weighted average that provides discount rates and multiples that they consider appropriate.

6 USING AN OPTION PRICING MODEL TO VALUE INVESTMENTS

Identifying the appropriate tracking portfolio to determine the certainty-equivalent cash flows is also an imperfect procedure, but no more so than the procedures that are used to identify appropriate valuation ratios and discount rates. We again need to stress that there are inherent uncertainties in valuing future cash flows whether we use the traditional DCF model (which focuses on the estimate of risk-adjusted discount rates) or the real options approach (which focuses on identifying a tracking portfolio), and both approaches require sound judgment on the part of the user.

Example—Valuing the Cotton Valley Investment Using the Binomial Option Pricing Model

In the Cotton Valley oil example, the financial options exactly matched the investment that was being evaluated. Unfortunately, as our discussion in the last section indicates, in even the best cases, there is likely to be some basis risk, and in other cases, there may be no traded option that even remotely resembles the option embedded in the investment being valued. As we will show in this section, when you cannot match an investment's cash flow with a traded option, you *may* be able to price the option component of the investment using information from forward prices. However, this requires an option pricing model, along with assumptions about the distribution of oil prices in the future.

In this section, we illustrate how to estimate the value of a call option on oil using a single-period **binomial option pricing** model, which makes the assumption that the future oil price follows a **binomial distribution** (see the Technical Insight box titled “Probability Trees and the Binomial Distribution”). In our reexamination of the Cotton Valley oil field investment, we make an assumption about the distribution of oil prices. Specifically, as we illustrate in Figure 4, we assume that oil prices in the next period will be equal to a high price of \$60.00/barrel or a low price of \$48.90/barrel.

Figure 4**Distribution of Year-End Oil Prices**
T E C H N I C A L
I N S I G H T
Probability Trees and the Binomial Distribution

The idea behind a probability tree is not new to most students of finance. Very simply, a *probability tree* specifies the outcomes of an uncertain event such as the price of a pound of copper at the end of next year and the probabilities associated with each of the possible prices. If there are only two possible outcomes, the probability tree is a binomial tree; if there are three possible outcomes, it is a trinomial tree; and so forth. For our purposes, we will stick to the simplest case—the binomial probability tree.

To illustrate the construction of a binomial tree, consider the binomial distribution of oil prices at the end of one year that are found in Figure 4. The high and low prices of oil at the end of one year are calculated in terms of the forward price of oil observed today for oil delivered at the end of Year 1, $F_{0,1}$.⁹

$$\text{High Price} = F_{0,1}u \text{ where } u = e^{+\sigma} \quad \text{and} \quad \text{Low Price} = F_{0,1}d \text{ where } d = e^{-\sigma}$$

To calculate these prices, we need the forward price of oil to be delivered in one year, $F_{0,1} = \$54.17$, and the standard deviation in annual oil price changes, $\sigma = .10232$. Therefore, $u = e^{+\sigma} = e^{.10233} = 1.10775$ and $d = .9027$, which implies that at the end of one year, the price of oil will either be

$$\text{High Price} = F_{0,1}u = \$54.17 \times 1.10775 = \$60.00$$

or

$$\text{Low Price} = F_{0,1}d = \$54.17 \times .9027 = \$48.90$$

The single-period binomial model can be extended to multiple periods with a few slight modifications. We demonstrate this procedure in Appendix B.

⁹ There are other ways to characterize the distribution of prices in a binomial lattice. However, we follow the approach taken by Robert L. McDonald, *Derivative Markets*, 2nd ed. (Boston: Addison Wesley, 2006).

TECHNICAL INSIGHT

Continuous Time Discounting

In the option literature, the standard convention is to use continuous time compounding/discounting. Consequently, the future value of \$1.00 compounded using continuous compounding at a rate of 10% for one year is equal to $e^{.10}$, which is equivalent to discrete time compounding using an annual rate of 10.517%, i.e., $(1 + .10517)$.

The first step in the investment valuation process is to identify the cash flows that accrue to the equity holder under each of the crude-oil-price scenarios. The cash flows equal the revenue generated by 5 million barrels of production sold for either \$60.00/barrel or \$48.90/barrel, minus the \$12/barrel cost of extracting the oil, plus the \$20.0 million due on the note. Thus, the summary information for the oil field investment is as follows.¹⁰

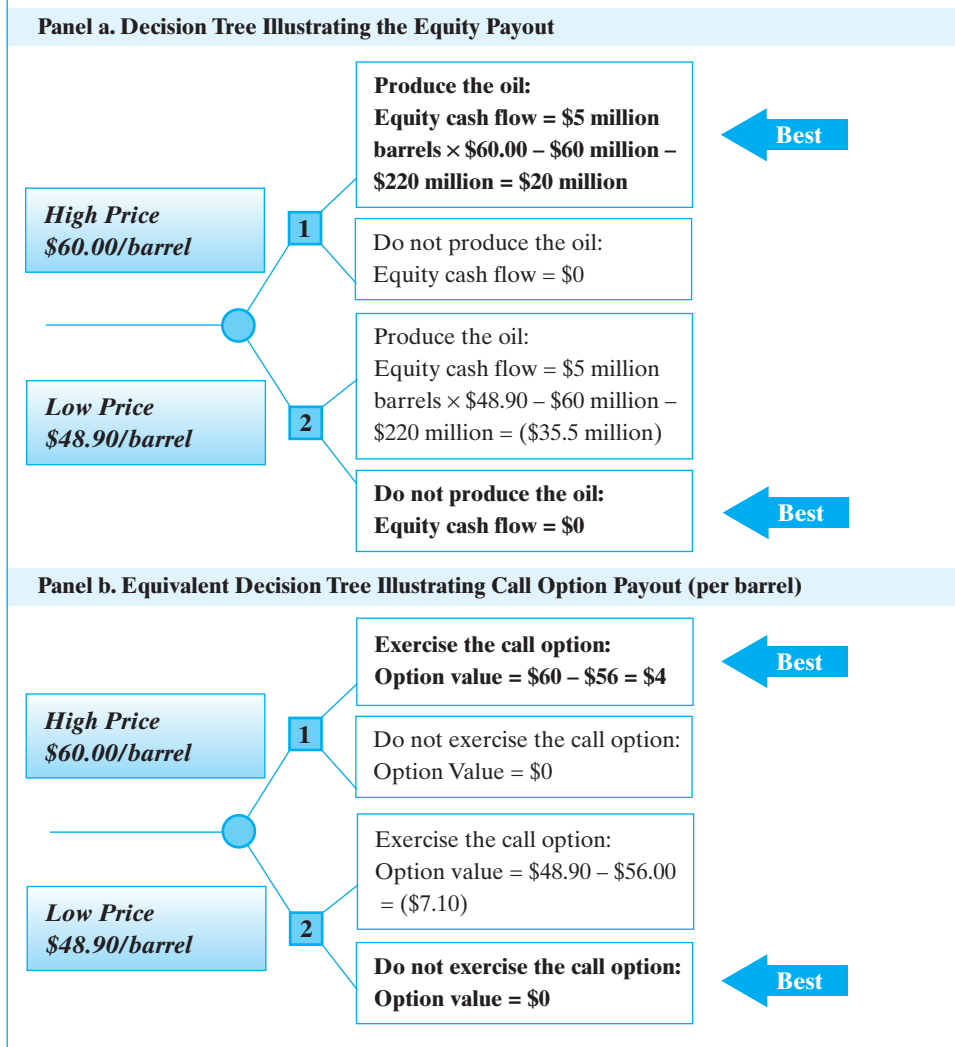
- The oil price next year will be either \$48.90 or \$60.00 per barrel, as illustrated in Figure 4.
- Extraction and transportation costs are known and equal \$12 per barrel.
- The risk-free interest rate is 8%.
- The forward price for oil observed today for delivery in one year is \$54.17/barrel.
- A total of 5 million barrels of oil can be extracted from the property, and the entire amount can be extracted next year.
- The oil field serves as the sole collateral for a \$200 million note that is due at the end of next year and that requires 10% interest compounded annually.
- The owner of the oil field has offered to sell his equity for \$5 million, and the buyer will assume the \$200 million note, which is secured by the property.

The question, then, is this: Is the offer a good one?

The decision trees found in Panels a and b of Figure 5 provide a convenient way to illustrate graphically how the decision process unfolds over time. The circle nodes represent uncertain events, such as different oil price possibilities, and the square nodes represent decision events—for example, whether or not to drill. The numbers found in the square nodes are used to keep track of the particular decision that is to be made as part of the decision problem (e.g., investment) being described.

Panel a of Figure 5 presents a detailed analysis of the cash flows that are realized from an uncertain future price of crude oil if the firm does not sell the oil in the forward market. If oil prices increase to \$60.00/barrel, the investment revenues will be \$300 million = 5 million barrels $\times$ \$60.00/barrel, leaving a profit of \$20 million after paying the extraction costs (\$60 million) and retiring the debt (\$220 million). If oil prices at year-end equal the low price of \$48.90/barrel, the investment will lose \$35.5 million after extraction costs and repayment of the debt. Because the loan is nonrecourse, the investor will default when oil prices are low, making the cash flow to equity equal to 0.

¹⁰ This is an existing note that has a \$200 million face value; however, given the assumptions in this example, its market value would be less.

Figure 5 Decision Tree for the Oil Production Problem

As Figure 5 illustrates, the payoff to the equity investors is identical to the payoff from holding a call option to buy 5 million barrels of oil with an exercise price of \$56/barrel. In both cases, the investment pays \$20 million (if oil prices turn out to be \$60/barrel) and nothing (if oil prices are \$48.90/barrel).

Again, it is worth noting that the traditional DCF approach would require (1) estimating the possible cash flow outcomes for the option in the high and low crude oil price states, (2) calculating the expected cash flow by weighting these cash flows using the probabilities of the high- and low-price states, and (3) then discounting the expected cash flow at an appropriate risk-adjusted discount rate. The alternative that is used in practice is to calculate the option's certainty-equivalent cash flow rather than its expected cash flow and to discount this value at the risk-free rate.

The advantage of this approach is that we do not need to know the *actual* probabilities of the two oil price scenarios. Instead, we calculate what is referred to as **risk-neutral probabilities**. Specifically, the *risk-neutral probabilities are hypothetical probabilities that make the forward price equal the expected price of oil next year*. We can solve for the risk-neutral probabilities using Equation 1 as follows:

$$\begin{aligned} \text{Forward Price for Oil}_{\text{Year } t} = & \left(\frac{\text{High}}{\text{Price for Oil}_{\text{Year } t}} \right) \times \left(\frac{\text{Risk-Neutral}}{\text{Probability}_{\text{High Price}}} \right) \\ & + \left(\frac{\text{Low}}{\text{Price for Oil}_{\text{Year } t}} \right) \times \left(\frac{\text{Risk-Neutral}}{\text{Probability}_{\text{Low Price}}} \right) \quad (1) \end{aligned}$$

Substituting values from our current example, we solve for p , which represents the risk-neutral probability that the \$60.00 price will prevail, as follows:

$$\$54.17 = p \times \$60.00 + (1 - p) \times \$48.90$$

where $(1 - p)$ is the corresponding risk-neutral probability that the \$48.90 price will prevail. By solving the above equation, we see that the risk-neutral probability of the high price occurring is 47.44%, which leaves a 52.56% risk-neutral probability for the low price.¹¹

Using the risk-neutral probabilities, we see that the risk-neutral expected value or certainty equivalence of the option payoff at its expiration date in one year is $.4744 \times \$4.00 + .5256 \times \$0.00 = \$1.90$. Discounting this certainty-equivalence payoff using the 8% risk-free rate provides an estimate of the value of the call option today. Thus, the value of one call option to acquire a barrel of oil via the investment just described is $\$1.75 = \$1.90e^{-.08}$, which is what we assumed earlier, and the value of an option to acquire 5 million barrels is \$8.75 million.¹²

How Does Volatility Affect Option Values?

How is the value of this equity position affected if uncertainty increases? In our discounted cash flow analysis, we learned that an increase in risk, other things remaining the same, led to an increase in the discount rate and a corresponding decrease in the present value of expected future cash flows. However, as we now illustrate, this is not necessarily the case for investments with option components.

Consider the case where the forward price stays at \$54.17/barrel, but the oil price when the economy is strong is increased to \$66.00/barrel, and the price when the economy

¹¹ As we discussed in Section 2, forward prices need not equal the expected value of future spot prices, because hedging pressure from market participants may cause forward prices to be either higher or lower than the expected future prices. This in turn implies that risk-neutral probabilities, which make the forward price equal to its certainty-equivalent value, will not generally equal actual probabilities. For example, if the forward price is lower than the expected price in the future, the risk-neutral probability of the high-price state will be lower than the actual probability, and the risk-neutral probability of the low-price state will be higher than the actual probability.

¹² The astute reader will notice that we have not presented anything that might be construed as an *option pricing model* in the traditional sense. Instead, we have valued the call option, simply using the certainty-equivalent DCF model that we have been using throughout this chapter. We were able to do this because we can observe forward prices for the underlying asset. We tackle situations in which forward prices are not observable.

is weak is decreased to \$44.50/barrel. In this case, the risk-neutral probabilities can be computed using Equation 1 to be 45% for the high-price state and 55% for the low-price state.¹³ Because the equity position has value only when the price of oil is \$66.00, the value of the equity in Year 1 is calculated as follows:

$$\frac{.45 \times (\$66/\text{barrel} \times 5 \text{ million barrels} - \$280 \text{ million})}{1.08} = \$20.833 \text{ million}$$

This is a substantial increase from the \$8.75 million value calculated with the lower volatility.¹⁴

This result may seem odd, because we normally associate greater uncertainty with higher risk, not higher value. However, the equity investor benefits from the increased volatility, which leads to the possibility of both higher and lower future oil prices, because higher *and* lower oil prices have an asymmetric effect on cash flows. Although higher oil prices always generate higher revenues, the loss associated with lower oil prices has a floor. Because the investor has the option to walk away from the investment, the investor cannot lose more than her original investment no matter how much oil prices decline.

Calibrating Option Pricing Models

It is often the case that traded options exist that resemble the embedded option being valued, but the basis risk is quite large. When this is the case, financial analysts infer information from the best available option and forward prices and use this information in combination with an *option pricing model* to value the options under consideration. Specifically, analysts use observed option prices to back out the volatility (or, equivalently, the spread between the price in the high and low states) of the price of the underlying commodity. For example, if the prices of traded options on oil are relatively high, they infer that oil prices are expected to be relatively volatile, which means that a higher volatility should be used to value the options embedded in an oil field. This process of using observed option prices to infer the parameters that are used for a model is known as *calibrating* the model. Appendix C contains an illustration of this process using the binomial option pricing model.

7 SUMMARY

The central point of this chapter is that the observable market prices of derivative contracts (forwards, futures, and options prices) provide information that can be used to value real investment projects. For example, forward prices provide the information needed to calculate certainty-equivalent cash flows, which can easily be valued by discounting them using the risk-free rate of interest. This approach should not be viewed as an alternative to DCF analysis. Indeed, a key learning point of this chapter is that observed prices of traded derivatives can be used to more accurately apply DCF analysis.

¹³ Where the forward price remains \$54.17, the risk-neutral probability of the high-price state, p , is calculated as follows:

$$\$54.17 = p \times \$66.00 + (1 - p) \times \$44.50$$

¹⁴ Most of the gain in value of the equity is due to a reduction in the value of the outstanding debt.

How useful are the methods described in this chapter for valuing real investment opportunities? In reality, it is unlikely that an investment's cash flows can be exactly replicated with publicly traded financial derivatives. However, this problem is no different from the problem of finding comparable firms for implementing relative valuation or the problem of estimating a project's risk-adjusted cost of capital. When this problem arises in a valuation analysis—and it almost always does, regardless of the method being used—financial analysts must exercise their professional judgment to come up with combinations of futures and/or options that track the investment cash flows as closely as possible.

In some cases, managerial judgment is required to select derivative prices for a traded commodity that is closely related to the commodity that underlies the project cash flows. For example, if an investment's cash flows are derived from the production and sale of heavy oil from Alberta, the analyst may use the West Texas crude oil futures price, adjusted for the typical \$4-per-barrel difference in prices of the two sources of crude. In other cases, more complex transformations are required. For example, if the terms of the traded option contracts do not match the specific attributes of the option embedded in the investment being valued, an option pricing model, such as the binomial option pricing model introduced in this chapter, can be used. To apply such a model, you should use information from the prices of traded options that have attributes that closely resemble the real options being valued.

As we mentioned at the outset of this chapter, we have illustrated the derivative valuation approach with examples of oil investments. The techniques that we have described were originally developed to value oil, gas, and other natural resource investments. However, the approach that we have described is currently being extended to value a wide range of investments in an expanding variety of industries.

PROBLEMS

1 USING DERIVATIVES TO ANALYZE A NATURAL GAS INVESTMENT Morrison Oil and Gas is faced with an interesting investment opportunity. The investment involves the exploration for a significant deposit of natural gas in southeastern Louisiana near Cameron. The area has long been known for its oil and gas production, and the new opportunity involves developing and producing 50 million cubic feet (MCF) of gas. Natural gas is currently trading around \$14.03 per MCF; the price next year, when the gas would be produced and sold, could be as high as \$18.16 or as low as \$12.17. Furthermore, the forward price of gas one year hence is currently \$14.87. If Morrison acquires the property, it will face a cost of \$4.00 per MCF to develop the gas.

The company trying to sell the gas field has a note of \$450 million on the property that requires repayment in one year plus 10% interest. If Morrison buys the property, it will have to assume this note and responsibility for repaying it. However, the note is nonrecourse; if the owner of the property decides not to develop the property in one year, the owner can simply transfer ownership of the property to the lender.

The property's current owner is a major oil company that is in the process of fighting off an attempted takeover; thus, it needs cash. The asking price for the equity in the property is \$50 million. The problem faced by Morrison's analysts is whether the equity is worth this amount. (Answer the following assuming zero taxes.)

- a. One possible response to the valuation question is to estimate the value of the project where the price risk of natural gas is eliminated through hedging. Estimate the value of the equity in the project where all the gas is sold forward at the \$14.87-per-MCF price. The risk-free rate of interest is currently 6%.
- b. Alternatively, Morrison could choose to wait a year to decide on developing it. By delaying, the firm chooses whether or not to develop the property based on the price per MCF at year-end. Analyze the value of the equity of the property under this scenario.
- c. The equity in the property is essentially a call option on 50 MCF of natural gas. Under the conditions stated in the problem, what is the value of a one-year call option on natural gas with an exercise price of 13.90 MCF worth today? (*Hint: Use the binomial option pricing model.*)

2 VALUING AN ENERGY INVESTMENT USING OIL OPTIONS The Pampa Oil Company operates oil and gas exploration throughout the panhandle of Texas. The firm was recently approached by a wildcatter named William “Wild Bill” Donovan with the prospect to develop what he thinks is a sure thing. Wild Bill owns the lease and wants to sell it to Pampa to meet some rather pressing gambling debts.

Wild Bill is extremely confident that there are 20,000 barrels of oil to be found, and he has engineering and geological reports to support his view. The value of the proposition hinges on the price of oil, the cost of exploration, and the cost of extracting the oil. Pampa Oil is very familiar with exploration and production in the area and is confident about its cost estimates. Pampa Oil estimates that the exploration would involve efforts expended over a period of six months and cost \$600,000 (which, for simplicity, we assume is paid at the end of the year). Pampa Oil also feels confident that the cost of extracting oil will be no more than \$8 a barrel. However, oil prices have been very volatile, and the experts in the economy predict that oil prices might hit \$50 a barrel by year-end or drop back to \$35, depending upon progress made in securing a lasting peace in the Middle East. Pampa Oil is, therefore, considering the possibility of deferring development of the oil field for six months. Waiting for six months will place Pampa Oil in a better position to determine whether or not to go ahead with exploration. The risk-free rate of interest is currently 5%, and the forward price of oil one year in the future is now trading at \$40 a barrel. What should the investment proposal be worth to Pampa? (You may assume a zero income tax rate.)

3 THE LOGIC OF HEDGING WITH OPTIONS Morrison Oil and Gas Company’s (from Problem 1) chief financial analyst is Samuel (Sam) Crawford. Sam completed his analysis, suggesting that the investment is indeed a good one for the company, and presented it to the firm’s executive committee. The executive committee consists of the firm’s CEO, CFO, and COO. The CFO thought Sam’s analysis was on target, but the COO and CEO were concerned about the fact that the hedging strategy would not work for the investment. In fact, they wondered why hedging the investment is such a good idea. Sam thought for a second or two before responding and decided how to best explain why the project is a good one and involves hedging the investment cash flows using one-year call options on natural gas. These call options, which have a \$13.90-per-MCF strike price, are selling for \$1.86 per MCF. Show how selling call options on 50 MCF of gas today and undertaking the investment provides Morrison with a hedged (i.e., risk-free) investment.

4 VALUING A COPPER MINING PROJECT USING FORWARD PRICES Harrington Explorations Inc. is interested in expanding its copper mining operations in Indonesia. The area has long been noted for its rich deposits of copper ore, and with copper prices at near-record levels, the company is considering an investment of \$60 million to open operations into a new vein of ore that was mapped by company geologists four years ago. The investment would be expensed (a combination of depreciation of capital equipment and depletion costs associated with using up the ore deposit) over five years toward a zero value. Because Harrington faces a corporate tax rate of 30%, the tax savings are significant.

The company's geologists also estimate that the ore will be of about the same purity as existing deposits, such that it will cost \$150 to mine and process a ton of ore containing roughly 15% pure copper. The company estimates that there are 75,000 tons of ore in the new vein that can be mined and processed over the next five years at a pace of 15,000 tons per year.

Harrington's CFO asked one of his financial analysts to come up with an estimate of the expected value of the investment using the forward price curve for copper as a guide to the value of future copper production. The forward price curve for the price per ton of copper spanning the next five years when the proposed investment would be in production is as follows:

2011	\$7,000/ton
2012	\$7,150/ton
2013	\$7,200/ton
2014	\$7,300/ton
2015	\$7,450/ton

In a study commissioned by the CFO last year, the firm's cost of capital was estimated to be 9.5%. The risk-free rate of interest on five-year Treasury bonds is currently 5.5%.

- Estimate the after-tax (certainty-equivalent) project free cash flows for the project over its five-year productive life.
- Using the certainty-equivalent valuation methodology, what is the NPV of the project?
- Assume now that the analyst estimates the NPV of the project using the certainty-equivalent methodology and it is negative. When the firm's CFO sees the results of the analysis, he suggests that something must be wrong because his own analysis using conventional methods (i.e., expected cash flows and the firm's weighted average cost of capital) produces a positive NPV of more than \$450,000. Specifically, he estimates that the price of copper for 2011 would indeed be \$7,000 per ton but that this would increase by 12% per year over the five-year life of the project. How should the analyst respond to the CFO's concerns?

5 VALUING A GOLD INVESTMENT OPPORTUNITY Jim Lytle, a financial advisor, recommends that his clients invest in gold. Specifically, he is advising a client to invest \$100,000 to purchase 175 ounces of gold bullion, with the expectation of holding the gold for a period of one year before selling it. The client points out that the futures price for 175 ounces of gold to be delivered in one year is \$104,000, which represents a 4% return on the \$100,000 investment, while the one-year Treasury yield is currently 5%. "Wouldn't it be better," the client asks, "if I sold 175 ounces of gold today for \$100,000, while purchasing

a forward contract to purchase the gold in one year at a price of \$104,000? I could then invest the \$100,000 in 5% Treasury bonds maturing in one year.”

- a. Analyze the returns for the investor’s proposed strategy. What is your recommendation?
- b. Why might the two alternatives offer different returns?

6 VALUING GOLD RECLAIMED FROM OLD PERSONAL COMPUTERS A number of industrial products include gold and silver as a component. Because they have very good conductive properties, the S&M Smelting Company engages in the recovery of gold from such products and is considering a contract to begin extracting the gold from the recycling of personal computers. The project involves contracting with the state governments of three midwestern states to dispose of their PCs. The project will last for five years, and the contract calls for the disposal of 200,000 PCs per year. Three tons of electronic scrap contains approximately one Troy ounce of gold. Moreover, each PC contains approximately 6 pounds of electronic scrap, and the processing cost involved in extracting the gold is \$67.50 per ton of scrap. In addition, the current (spot) price of gold is \$592.80, and the forward price curve for the price per ounce of gold spanning the next five years is as follows:

2011	\$679.40/ounce
2012	\$715.10/ounce
2013	\$750.60/ounce
2014	\$786.90/ounce
2015	\$822.80/ounce

S&M estimates that the firm’s cost of capital is 10.5%, and the risk-free rate of interest on five-year Treasury bonds is currently 5.0%. In addition, S&M faces a 30% tax rate, and the entire investment of \$450,000 made in the project in 2010 will be depreciated using straight-line depreciation over five years with a zero salvage value.

- a. Estimate the after-tax (certainty-equivalent) project free cash flows for the project over its five-year productive life.
- b. Using the certainty-equivalent valuation methodology, estimate the NPV of the project.
- c. If we assume that gold prices will increase at a rate of 7% per year over the next five years, what is the NPV of the project using the traditional WACC method of analysis based on expected project free cash flows, where the WACC is estimated to be 10.5%? What rate of growth in gold prices is required to produce the same NPV using the traditional WACC approach as with the certainty-equivalent approach used in part b?

Option Basics—A Quick Review

This appendix is targeted toward the reader whose knowledge of options and their payoffs is either limited or just rusty from years of nonuse. Our objective is to develop a platform from which to build an understanding of the use of options throughout this chapter and the two that follow.

What Is an Option?

An **option** gives its owner the right, but not the obligation, to buy or sell a specified asset at a contractually agreed-upon price (the option's *exercise price* or *strike price*) within a specified period of time. A **call option** gives the owner the right (but, again, not the obligation) to buy the asset at the exercise price. A **put option** gives the owner the right (but, again, not the obligation) to sell the asset at the exercise price. In either case, the option owner can walk away if it is not in his or her best interest to buy or sell.

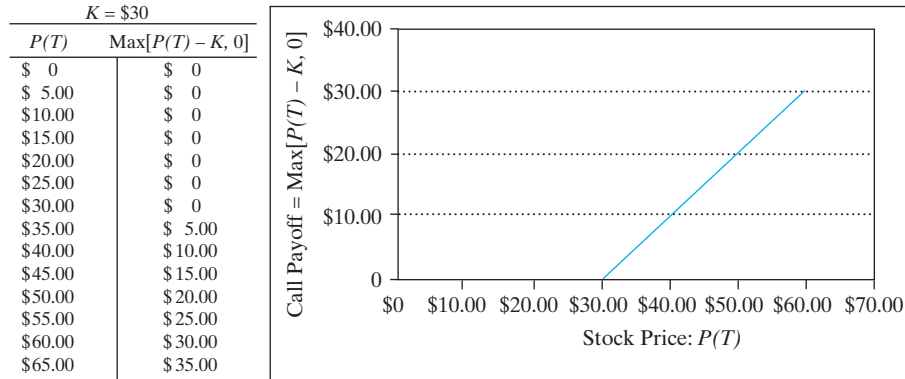
Option contracts come in two basic types that differ in *when* they can be exercised: The **European option** provides the holder the right to exercise the option *only* on its expiration date; the **American option** allows the holder to exercise it at *any time* up until the expiration date of the option.

How Can We Characterize the Payoffs to Option Contracts?

Consider the economic consequences of owning a call option on a share of stock. The payoff to the holder of the option at maturity (i.e., date T when the option expires) can be represented graphically using the position diagram in Figure A-1. The expiration-date payoff to owning the call option (or being long in the call) is represented by the 45° line moving up and to the right, beginning at the exercise price, K , of \$30. If the stock price, $P(T)$, is, say, \$50 and the exercise price, K , is \$30, then on the day the option expires, it has a value of $P(T) - K = \$50 - 30 = \20 . However, if, on the expiration date of the option, the stock price falls below \$30, the option is worthless. For example, if the stock price on the expiration day were only \$25, then the holder would not exercise the option, because it has no value. (Who would pay \$30 to exercise an option to buy a share of stock that one could purchase in the market for only \$25?) Thus, we want to own call options when we expect that prices of the underlying asset are going to increase.

Defining Call Option Payoffs

Analytically, we write the payoff to the call option as $\max[P(T) - K, 0]$, which is read “the maximum of $P(T) - K$ or zero.” When the option payoff is positive (i.e., $P(T) > K$),

Figure A-1**Expiration-Date Payoff to a Call Option
[Exercise Price (K) = \$30]**

the option is said to be “in the money.” Similarly, when $P(T) = K$, the option is “at the money,” and when $P(T) < K$, the option is “out of the money.” Consequently, for a call option there are two critical states at expiration—the stock price is either greater than the exercise price or it is not—that are summarized in Equation A.1:

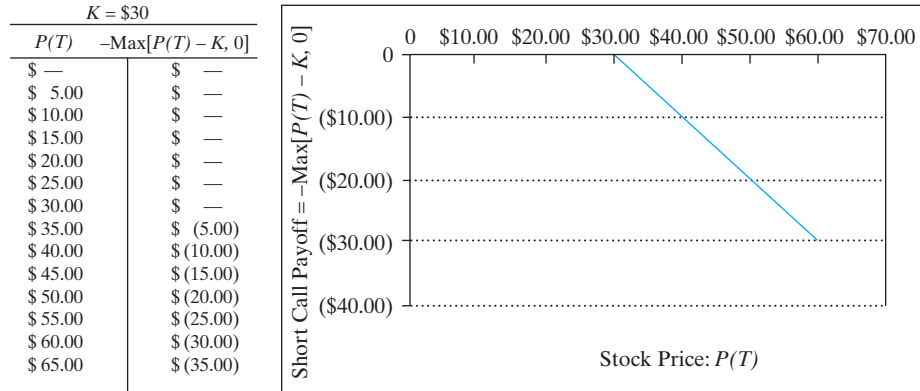
$$\begin{array}{l} \text{Call Option} \\ \text{Payoff at} \\ \text{Expiration } (T) \end{array} = \text{Max}[P(T) - K, 0] = \begin{cases} P(T) - K & \text{if } P(T) > K \\ 0 & \text{if } P(T) \leq K \end{cases} \quad (\text{A.1})$$

If $P(T) > K$, then the call option has value equal to $P(T) - K$. Otherwise, the call expires worthless.

Figure A-1 represents the expiration-day payoff to *the owner of a call option* with an exercise price, K , of \$30. What does the payoff look like for the *person who sold the call option*? The answer is straightforward. For every dollar the owner of the call option makes, the seller loses a dollar. For example, if the price of the stock is \$50, the owner of the call owns a security that is worth \$20. (She can purchase a share of stock worth \$50 by exercising the call option for \$30.) However, the issuer of the call option is obligated to deliver the share of stock worth \$50 in the market for the exercise price of \$30 (a loss of \$20). Therefore, the payoff to selling (shorting, which is referred to as *writing*) the call option, shown in Figure A-2, is the mirror, or inverse, image of Figure A-1. Analytically, the payoff to shorting a call is the negative of the payoff to owning (being “long in”) the call—that is, $-\text{max}[P(T) - K, 0]$.

Defining Put Option Payoffs

A put option gives its owner the right, but not the obligation, to sell an asset at a prescribed exercise price and within a specified time period. Thus, we want to own put options when we expect that prices of the underlying asset are going to fall. This is because the put option has value when the price of the underlying asset falls below the stated exercise price.

Figure A-2**Expiration-Date Payoff to the Seller of a Call Option
[Exercise Price (K) = \$30]**

Consider the put option payouts in Figure A-3, where the put option has an exercise price of \$30. If the price of the underlying asset on which the put is written has a price of \$25, then the holder of the put can exercise the option to sell the stock for the \$30 exercise price. However, if the price of the stock is \$40, the put will expire worthless, because the stock can be sold in the market for \$40, while the put exercise price is only \$30.

You can think of a put option contract like an insurance policy. That is, in the event that the value of the underlying asset drops below some threshold level, we are protected because the payoff to the put increases dollar for dollar with any further decline in the value of the underlying asset. Similarly, in the event your home burns down, a homeowner's insurance policy provides you with the right to sell the home to the insurance company for the value of the insured damage.

Analytically, we can express the payoff to owning the put option (a long position) as $\text{max}[K - P(T), 0]$. That is, the put has positive value equal to $[K - P(T)]$ only when the

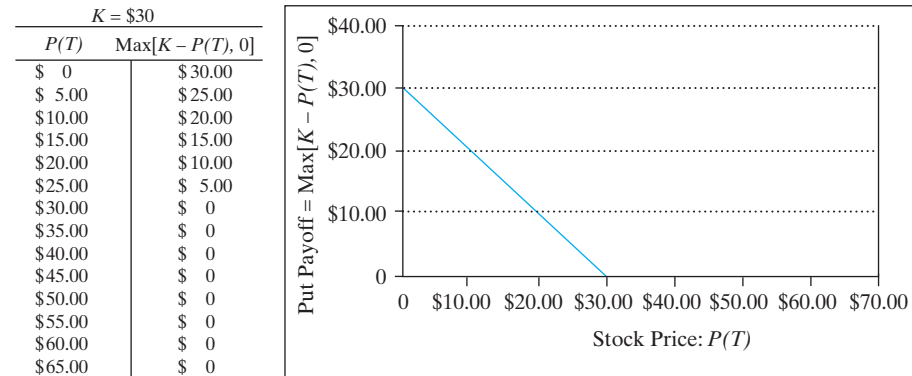
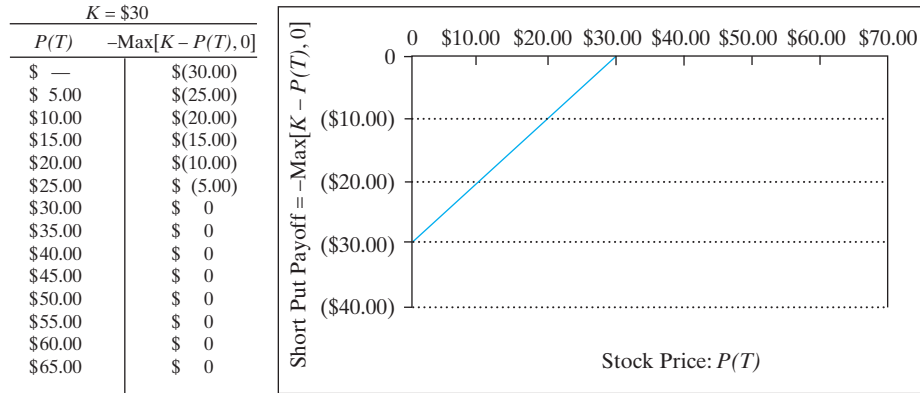
Figure A-3**Expiration-Date Payoff to the Owner of a Put Option
[Exercise Price (K) = \$30]**

Figure A-4**Expiration-Date Payoff to the Seller of a Put Option
[Exercise Price (K) = \$30]**

price of the stock falls below the exercise price, K . Equation A.2 captures the payoff structure of the put option on date T :

$$\begin{array}{l} \text{Put Option} \\ \text{Payoff at} \\ \text{Expiration } (T) \end{array} = \text{Max}[K - P(T), 0] = \begin{cases} K - P(T) & \text{if } P(T) \leq K \\ 0 & \text{if } P(T) > K \end{cases} \quad (\text{A.2})$$

Once again, the payoff to the seller of the put option is the mirror (or inverse) image of the payoff to the one who buys it (i.e., the owner), as Figure A-4 illustrates.

If the stock price is \$25, the put option is worth \$5 to the holder of the option. Correspondingly, the person who sold the put (commonly referred to as the *put writer*) is obligated to pay the exercise price of \$30 for the share of stock that is worth only \$25 in the market and consequently faces a loss of \$5.

Risk and the Valuation of Options

An important feature of most option contracts is that their value is higher when the underlying asset becomes more volatile or risky. For example, stock options are more valuable when the volatility of the stock is higher. The intuition that supports this statement is straightforward. For most investments, volatility is viewed negatively, because the increased probability of achieving very high returns is offset by a higher probability that very negative returns will be realized. However, the expected payoff of an option generally increases as the volatility of the underlying investment increases. This is because higher volatility increases the probability that the option payoff will be very high, but because the maximum loss on an option contract is fixed, the increase in the upside potential is not completely offset by increases in downside risk.

Valuing Call Options Using the Black-Scholes Model

To this point, we have restricted our discussion to the expiration-date payoff of the option, or the value of the option on the day it expires. Valuing options prior to the expiration date provides a much more challenging problem, and perhaps the best-known valuation

model is the one developed by Fisher Black and Myron Scholes. Their model can be used to value options that can be exercised only at expiration (i.e., European options). Another significant difference between their model and the binomial option pricing model that we discuss in Appendix C of this chapter is that the distribution of stock prices in the Black-Scholes model is assumed to be continuous rather than binomial. However, the basic approach to the derivation of the pricing model is the same. By buying shares of stock and simultaneously selling options on the stock, the investor can create a risk-free payoff. The resulting option pricing equation for a call option on a stock whose current price is P_{Today} and that expires in period T can be written as follows:

$$\text{Call}(P, T, K) = P_{\text{Today}} N(d_1) - Ke^{-r_f T} N(d_2) \quad (\text{A.3})$$

where K = the option's strike price, r_f = the risk-free rate of interest,

$$d_1 = \frac{\ln\left(\frac{P_{\text{Today}}}{K}\right) + \left(r_f + \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}}, \quad \text{and} \quad d_2 = d_1 - \sigma\sqrt{T}$$

$N(d_i)$ is the probability that a value less than d could occur under the standard normal distribution (i.e., mean zero and standard deviation equal to 1); σ^2 is the annualized variance in the returns (continuously compounded) on the stock, i.e., $\ln[P(t+1)/P(t)]$; and all the remaining terms retain their previous definitions.

To illustrate the use of the Black-Scholes formula, let us consider the following example: The current price of the stock on which the call option is written is $P_{\text{Today}} = \$32.00$; the exercise price of the call option $K = \$30.00$; the maturity of the option is $T = 90$ days or .25 years; the (annualized) variance in the returns of the stock is $\sigma^2 = .16$; and the risk-free rate of interest is $r_f = 12\%$ per annum. To estimate the value of the call option using Equation A.3, we must first solve for d_1 and d_2 as follows:

$$d_1 = \frac{\ln\left(\frac{P_{\text{Today}}}{K}\right) + \left(r_f + \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}} = .572693$$

and

$$d_2 = .1432 - .4\sqrt{.25} = .372693$$

Now, using Equation A.3 and the table of areas under the standard normal distribution (found in any standard statistics book), we calculate the value of the option by substituting the appropriate values into Equation A.3, i.e.,¹⁵

$$\begin{aligned} \text{Call}(P, T, K) &= P_{\text{Today}} N(d_1) - Ke^{-r_f T} N(d_2) \\ \text{Call}(P, K, T) &= \$32 (.716574) - \$30 e^{-.12(.25)} (.645311) = \$4.1437 \end{aligned}$$

Note that the option's value is \$4.14 even though the current \$32 price of the stock is only \$2.00 greater than the exercise price of \$30. This premium arises because there is the potential for the value of the call option to go up even higher if the stock price should rise over the next 90 days. In fact, if the term to maturity of the option was six months rather than three months, the value of the option would be \$5.58.

¹⁵ We can use the Normsdist (d_1) function in Excel to calculate $N(d_1)$ directly.

Multiperiod Probability Trees and Lattices

In the chapter, we introduced the binomial tree to characterize the outcomes of uncertain future oil prices. In this appendix, we extend the single-period tree to multiple periods using forward prices. There are multiple ways to construct a binomial tree, and the method we describe here develops a forward tree.¹⁶

Figure B-1 produces a three-year binomial lattice for the price of crude oil with a volatility (σ) estimate of .10 for the annual changes in oil prices. We calculate the high and low prices for Year 1 based on today's forward price for oil to be delivered at the end of Year 1 ($F_{0,1}$) as follows:

$$\text{High Price (for Year 1)} = F_{0,1} u$$

where $u = e^{+\sigma}$ and σ is the volatility in oil price changes, and

$$\text{Low Price (for Year 1)} = F_{0,1} d$$

where $d = e^{-\sigma}$.

Given a \$52.04-per-barrel forward price for crude oil at the end of Year 1 and an estimated volatility in annual price changes (σ) equal to .10, we calculate the high and low prices for Year 1 as follows:

$$\text{High Price (for Year 1)} = F_{0,1} u = \$52.04 e^{.10} = \$57.51$$

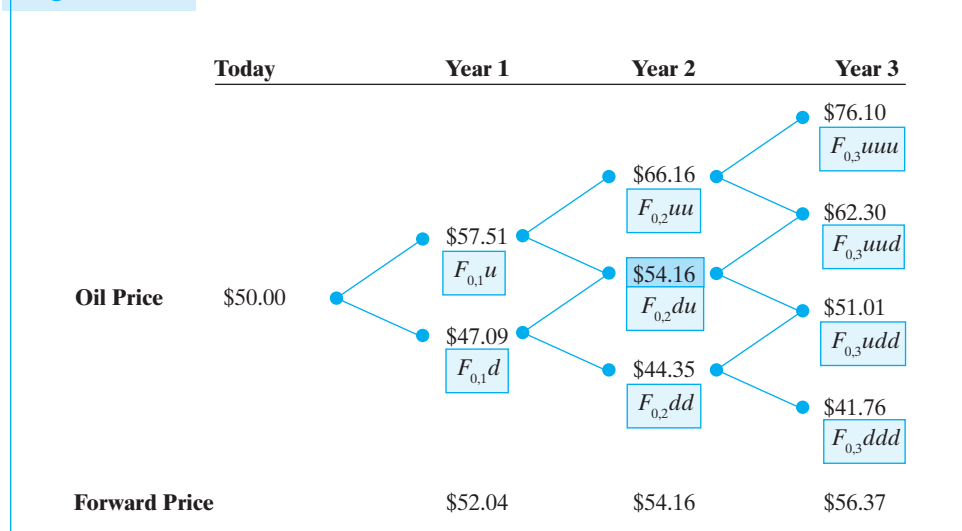
and

$$\text{Low Price (for Year 1)} = F_{0,1} d = \$52.04 e^{-.10} = \$47.09$$

The formulas used to calculate the possible prices in the binomial lattice for Years 2 and 3 are based on the forward prices for each year and are found just below the price in each node of the lattice.

The binomial tree found in Figure B-1 is a recombining binomial tree or lattice. The practical importance of this is that the resulting tree utilizes fewer price nodes. For example, consider the middle node found in Year 2 of Figure B-1, equal to \$54.16. This price can be achieved by a sequence of one high price in Year 1 followed by a low price in Year 2, or a low price in Year 1 followed by a high price in Year 2. This shared price-node feature of the recombining binomial tree or lattice means that there are fewer price nodes to deal with in the tree.

¹⁶ See Robert McDonald, *Derivative Markets*, 2nd ed. (Boston: Addison Wesley, 2006).

Figure B-1**Binomial Distribution for the Price of Crude Oil****Summing Up**

We close our brief overview of the construction of the binomial lattice with a few key observations:

- First, we construct a special form of binomial tree, known as a recombining tree or lattice, in which up moves and down moves are restricted to be symmetrical. This restriction facilitates the computation of large multiperiod trees.
- Second, the source of uncertainty in future values contained in the lattice is determined by the underlying volatility in the annual price changes.
- Third, the technique we use to construct the binomial lattice is designed to be consistent with observed forward prices.
- Finally, although the examples we use in this chapter all use annual periods, the length of each time step is not restricted to one year. We simply adjust the risk-free rate and annualized volatility to correspond to the length of one *time step* in the lattice. Note that by using multiple time steps per year, we greatly expand the number of data points (value estimates) per year, thereby improving the precision with which we are able to replicate the distribution of end-of-year values.

Calibrating the Binomial Option Pricing Model

This chapter introduces a valuation approach that employs information from the financial derivatives markets to value real investment opportunities.

This calibration and valuation process involves three steps:

- Step 1:** Select an option pricing model that provides a reasonable description of option prices. For our purposes, we will use the binomial model.
- Step 2:** Calibrate the option pricing model by determining which model parameters do the best job of describing observed option prices. Most of the model parameters, including contract terms and the risk-free interest rate, are observable. The key unobservable variable is the volatility of the future commodity prices. In the binomial model, this is the risk-neutral probabilities of the high and low states. The volatility that best explains observed option prices is generally referred to as the volatility that is implied by observed option prices (i.e., the **implied volatility**).
- Step 3:** Finally, we substitute the contract terms for the option we are trying to value, along with the implied volatility from the observable options, into the binomial option pricing model and estimate the value of the option.

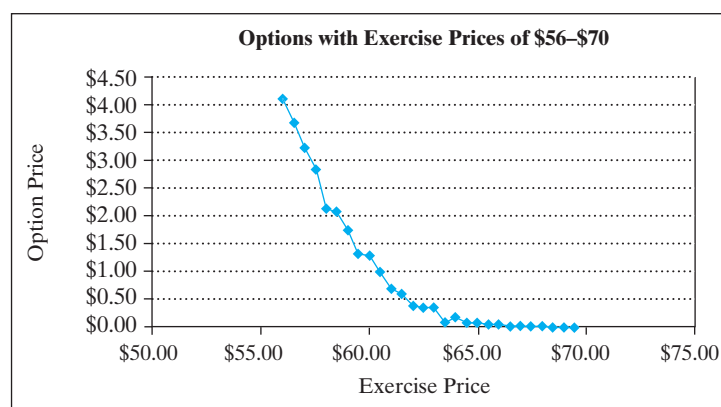
To illustrate how this calibration to observed market prices can be done, consider the problem we face if we need to know the value of a call option on a barrel of crude with one year to maturity and an exercise price of \$52. If options with exercise prices of \$50 and \$54 were publicly traded, you might be tempted to simply interpolate these option values and use an average of the two observed values as our estimate of the value of the \$52 option. However, as the Technical Insight box on next page illustrates, the relationship between exercise prices and option value is not linear, and such an interpolation could provide a very poor estimate of the value of the option with a \$52 exercise price.

Figure C-1 contains the binomial distribution of prices used to derive this option value. What we want to do now is to use the information reflected in the valuation of the option with the \$56 exercise price to value the new option.

Panels b and c of Figure C-1 describe the option valuation process for the single-period binomial option pricing model. In Panel b, we define the call option payoff for each of the two price states to be \$8 [=Max(\$60 – 52, 0)] for the high-price state and \$0 [=Max(\$48.90 – 52, 0)] for the low-price state. We would like to estimate the value of a call option with an exercise price of \$52. However, what we can observe in the option market is a call option for crude oil with a one-year maturity and a \$56 exercise price that sells for \$1.75.

Option Values and Exercise Prices

When trying to value a call option with an exercise price that is not observed in the option market, we may be tempted to try to simply interpolate a price using observed market prices and their exercise prices. Unfortunately, interpolation (which involves the use of a linear approximation) does not work well with option prices. Because, as the following graph indicates, the relationship between option values and exercise prices is not linear:



The option prices represent call options on crude oil, with exercise prices ranging from \$56.00 to \$70.00 at a time when the price of crude was around \$55.00 a barrel. The nonlinearity in option prices is very pronounced, suggesting that any use of linear interpolation will be fraught with estimation error.

Solving for the implied volatility, we estimate that $\sigma = .10232$. We use the implied volatility to calculate the risk-neutral probability of the high price of oil as follows:

$$P = \frac{e^r - e^{r-\sigma}}{e^{r+\sigma} - e^{r-\sigma}} = 0.4744$$

We are now ready for Step 3 (see Panel c in Figure C-1) in the three-step calibration process. Using the \$52 exercise price, and the risk-neutral probability of .4744, we get the option price estimate of \$3.51.

In this brief appendix, we have demonstrated how observed market price information can be incorporated into an option pricing model to calibrate a model that can be used to value options that are not publicly traded. The simple example we used involved

Figure C-1 Using the Binomial Option Pricing Model

Panel a. The Price Distribution for Crude Oil in One Year



Panel b. Option Valuation Using the Binomial Option Pricing Model

	Price of Crude Oil	Call Option Payoff	Risk-Neutral Probability*
High-Price State	\$60.00/barrel	Max (\$60 – 52, 0) = \$8	.4744
Low-Price State	\$48.90/barrel	Max (\$48.90 – 52, 0) = 0	.5256

*The forward price of crude oil for the end of Year 1 is equal to the current price, \$50.00, compounded one year at the risk-free rate of 8% or $\$50e^{.08} = \54.17 . Risk-neutral probabilities are the probabilities that we apply to both the high- and low-price states for crude to obtain an average equal to the forward price of crude oil.

Panel c. Calculate the Value of the Call Option

Call Option (Exercise Price = \$52.00, 1 Year to Maturity)

$$\begin{aligned}
 &= (\$8.00 \times .4744 + \$0.00 \times .5256)e^{-.08} \\
 &= (\$3.80 \times .9231 = \$3.51)
 \end{aligned}$$

the estimation of an option with an exercise price that is not observed in the market. However, we could also have estimated the value of an option with a maturity not represented in the market or even estimated the value of an option for a particular commodity for which options are not traded but that is closely related to another commodity whose option prices are available.

